

BEST PRACTICES FOR ERASING DATA

Before retiring or reallocating a mobile device, implementing a robust data sanitization and decommissioning process is critical to prevent unauthorized recovery of sensitive information. A study by Wired found that even after factory resets on unlocked Android devices, residual data can be retrieved with inexpensive tools, underscoring the need for encryption-first wipes and secure overwrites. The NIST SP 800-88r1 “Clear, Purge, Destroy” guidelines recommend layered erasure methods – logical overwrite followed by physical destruction where necessary – to meet corporate and regulatory standards.

Pre-erase preparation

1. **Inventory and classification:** Log device identifier (IMEI/serial), owner, and data classification tier (Tier 1–3) in your asset disposal register.
2. **Revocation of credentials:** Remove MDM profiles and revoke any VPN or certificate-based credentials from enterprise identity platforms to sever device-to-network trust.

Logical erasure techniques

1. **Encrypt before reset:** On Android, enable full-disk encryption; on iOS, set a strong passcode so that a subsequent factory reset cryptographically renders residual data inaccessible.

2. Factory reset with overwrite:

- **Android:** After encryption, perform Settings → System → Reset options → Erase all data (factory reset) and then use an overwrite tool (e.g., Android's shred command in a custom recovery) to write random data over free space.
- **iOS:** Use Settings → General → Reset → Erase All Content and Settings; Apple's secure erase leverages hardware-rooted key destruction for complete sanitization.

Remote wipe considerations

Remote wipe is invaluable for lost or stolen devices, but it depends on device power and network availability, and may leave firmware intact – allowing network re-entry if attackers disable wiping commands. Always follow up with physical sanitization when possible.

When devices cannot be logically sanitized – due to non-standard storage or high-value data – physical destruction is necessary. NIST 800-88r1 endorses shredding, degaussing, or disintegration of memory chips to ensure irrecoverability. E-waste specialists can provide certified destruction and chain-of-custody documentation to satisfy compliance. It's important to follow these steps for erasing data thoroughly.

including keychain and Health data.

Google One backup (Android / cross-platform)

1. Activate Backup

- Settings → Google → Backup → Back up now.

2. Contents

- App data, call history, contacts, device settings, SMS (Android), and 15 GB of Drive/Photos.

3. Upgrade Options

- 100 GB–30 TB plans; family sharing available.

Pro Tip: Use Google Photos' Locked Folder for sensitive images, requiring device authentication to view.

Testing and recovery procedures

A backup untested is a risk. Establish routine drills.

1. Verify integrity

- **For local:** Mount or restore a sample file from Finder/iTunes

or ADB-extracted archive.

- **For cloud:** Download a known backup version, decrypt, and open key documents.

2. Full restore drill

- Quarterly, perform a full device restore:
 - **iOS:** Erase device → Restore from encrypted backup.
 - **Android:** Factory reset → `adb restore mobile_backup.ab`.

3. Selective restore workflow

- Teach users to recover individual contacts, photos, or app settings without full wipe.

4. Document procedures

- Maintain runbooks with step-by-step restore commands and credential locations.

Example: After a ransomware test on a lab device, the team successfully restored corporate email profiles from backup in under 10 minutes, validating SLAs for critical system recovery. 